Roberto Gallego Barbarán

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EDUCATION AND SPECIALIZED TRAINING

University of Málaga 2022 – 2026
BSc in Software Engineering Málaga, Spain

University Extension in Blockchain Technologies (490h) Feb. 2026 – Jun. 2026
University of Málaga (UMA) Málaga, Spain

EXPERIENCE

DEKRA Sept. 2026 – Mar. 2027
AI & Automation Engineer Intern Málaga, Spain

- Automated laboratory-results analysis with AI agents in Python, LangChain and LangGraph, with RAG over vector databases, reducing manual processing and improving technical-data lookup.
- Designed pipelines connecting LLMs to external tools via MCP (Model Context Protocol) and orchestrating decision flows with LangGraph state graphs, removing repetitive internal tasks.

IOBuilders (University-Industry Collaboration) 2026
Backend & Blockchain Developer Málaga, Spain

- Delivered a tokenized bond platform: Go backend and Daml contracts, multi-service infrastructure with Docker Compose and Nginx, and Playwright end-to-end tests for issuance, transfer and burn flows.
- Kept Go business logic and the Canton Ledger consistent via gRPC, with automated tests across the token lifecycle.

PROJECTS

Intelligent Game Validation | *TypeScript, Node.js, Playwright, OCR, GitHub Actions* | GitHub 2026

- 4th place at Hackathon UMA 2026 (Betsson challenge, team of four) with a TypeScript/Playwright tool that validates casino games in cross-origin canvas iframes as the player sees them, with no DOM access.
- Computer-vision pipeline (pixelmatch, template matching, tesseract.js OCR) cross-checked with provider traffic; CI in GitHub Actions with 42 unit tests and a fault-injection negative control.

Bancosol Campaigns | *Java 17, Spring Boot, JPA, Spring Security, JWT, MariaDB* | GitHub 2026

- Architected the campaign management module (Spring MVC, services, JPA repositories, DTOs) with RBAC via Spring Security and JWT, and MariaDB persistence for stores, shifts, volunteers and incidents, tested with JUnit and Mockito.

Cava Agent | *Python, smolagents, Gradio, pytest, GitHub Actions* | GitHub 2026

- Turned a fixed LLM pipeline into a ReAct agent (smolagents) that picks its own tools (Yahoo Finance, SMA50/SMA200 trend, news, weather) with a swappable LLM (Groq, Gemini).
- CI/CD in GitHub Actions: ruff and pytest on Python 3.10–3.12, deploying the Gradio demo to Hugging Face Spaces after CI passes; daily Telegram bulletin via cron.

TECHNICAL SKILLS

AI & ML: LangChain, LangGraph, MCP, RAG, vector databases, LLM agents, smolagents, computer vision/OCR.
Backend & APIs: Java 17, Spring Boot/MVC, JPA, Spring Security/JWT, Maven, SQL, MariaDB/MySQL, REST APIs.
DevOps & Infrastructure: Docker/Compose, Kubernetes, GitHub Actions, Linux, Nginx, CI/CD, Terraform/IaC basics.
Languages & tools: Python, Go, TypeScript, JavaScript, C/C++, Git, JUnit/Mockito, pytest, Playwright.
Blockchain & distributed systems: Solidity, Daml, Canton, Smart Contracts, gRPC.
Spoken languages: Spanish native, English C1.